ARTIFICIAL INTELLIGENCE IN MENTAL HEALTHCARE

A PROJECT REPORT

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MAY 2026

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BONAFIDE CERTIFICATE

Certified that this project report "KEFFI AI – A MENTAL HEALTH CHATBOT" is the Bonafide work of MADHUMATHI S (422623104003), BALAJI P (422623104035), MALINI V (422623104048) who carried out their project under my supervision.

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EXAMINED ON: 05/08/2026 Afternoon Session (AN) | Panel 2 | Chennai Region Venue

INTERNAL EXAMINER EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We express our deepest gratitude to our project guide Dr. S. Sivanesh, Assistant Professor and Head of Department of Computer Science and Engineering, University College of Engineering Panruti, for his invaluable guidance, continuous encouragement, and constant support throughout the development of this project.

We extend our sincere thanks to Anna University and TNSDC Naan Mudhalvan Niral Thiruvizha 2025-2026 team for providing us the opportunity and platform to present our work in digital health tech.

ABSTRACT

Mental healthcare accessibility remains a major global challenge due to high therapy costs, limited availability of psychiatrists, and social stigma. While conventional psychotherapy provides structured support, treatment is usually restricted to one hour per week, leaving patients unmonitored during vulnerable moments throughout the rest of the week. Existing self-help chatbots attempt to address this gap, but most rely on basic rule-based scripts that lack long-term conversational memory, emotional personalization, and crisis safety protocols.

To address these challenges, this project introduces Keffi AI, a clinical digital therapeutics platform developed to assist patients dealing with anxiety, depression, and stress. The system utilizes a fine-tuned BERT transformer model to classify user input into 96 distinct emotional states. To maintain conversational context across sessions, the backend combines relational database storage running in Write-Ahead Logging mode with a vector memory engine, storing user history securely by patient identifier.

Patient condition is monitored using a dynamic Mental Health Quotient score alongside an acoustic voice prosody analyzer that processes speech pitch and energy. For users experiencing acute panic, a hands-free voice feature enables natural spoken interaction. System reliability is supported through a dual-model processing cascade, while diagnostic decisions are explained using feature attribution methods. In crisis situations, an automated workflow triggers immediate alerts and helpline information.

By combining a patient sanctuary interface with a clinician dashboard, Keffi AI offers a practical digital tool connecting self-guided recovery with professional psychiatric care.

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CHAPTER 1: INTRODUCTION

1.1 Overview of Mental Healthcare

Mental healthcare accessibility remains a critical global challenge. Millions of individuals suffer from major depressive disorder, generalized anxiety, and acute stress without access to timely intervention. High consultation fees, shortage of accredited psychiatrists, and societal stigma prevent individuals from seeking professional therapeutic care.

1.2 The 167-Hour Gap & Need for Continuous Monitoring

Standard psychotherapy sessions occur once a week for 1 hour. The remaining 167 hours represent an unmonitored window where negative cognitive distortions escalate undetected. Keffi AI bridges this gap by offering continuous digital therapeutics monitoring.

1.3 Proposed System Architecture

Keffi AI is built as a clinical digital therapeutics platform combining fine-tuned BERT transformer multi-emotion classification, vector memory persistence, dynamic Mental Health Quotient scoring, 7-mode therapeutic interventions, and clinical admin oversight.

1.4 Therapeutic Support System

The platform integrates evidence-based psychological frameworks including Cognitive Behavioral Therapy reframing, Acceptance and Commitment Therapy, Socratic reflective questioning, 4-7-8 breathing, and 5-4-3-2-1 grounding exercises.

1.5 Objective of the Project

The primary objective of Keffi AI is to deliver continuous, emotionally intelligent, crisis-safe digital therapeutics support, preventing patient treatment attrition and supporting long-term psychiatric recovery.

1.6 Hands-Free Real-Time Voice-to-Voice AI Architecture

In addition to text-based interaction, Keffi AI incorporates a hands-free real-time Voice-to-Voice AI architecture. Patients experiencing acute anxiety, motor fatigue, or panic often find typing difficult. Keffi AI utilizes browser-native speech recognition for real-time speech-to-text transcription and speech synthesis for voice output. When a patient speaks into the microphone, Keffi AI captures the spoken utterance, processes the emotional context through the language model cascade, and speaks back Keffi's therapeutic reply out loud in a warm, gentle voice, creating an accessible, voice-first digital therapy session.

CHAPTER 2: LITERATURE SURVEY

2.1 Deep Learning Emotion Detection

Kumar & Singh (2021) established that bidirectional transformer models process contextual text symmetrically, fine-tuning over 27 GoEmotions categories to classify complex emotional states beyond binary sentiment.

2.2 Sentiment Analysis for Mental Health Monitoring

Smith et al. (2020) demonstrated sentiment classification across social media dialogues to detect early risk signs of loneliness and emotional burnout.

2.3 AI Chatbots for Mental Health Support

Johnson (2019) evaluated early rule-based chatbots, identifying critical limitations: lack of long-term memory, generic responses, and absence of clinical safety fallback.

2.4 Detecting Depression from Text

Sharma et al. (2022) utilized sequential text models over mental health datasets to identify hopelessness, emotional withdrawal, and depressive language patterns.

2.5 Multi-Emotion Classification using Transformers

Verma (2023) validated multi-label RoBERTa and BERT transformers for fine-grained emotional category identification.

2.6 Multi-LLM 70B Engine Cascade & High Availability

To ensure zero downtime and sub-second response delivery, Keffi AI implements a multi-model cascade engine. The primary inference engine utilizes high-speed hardware processing for sub-500ms response generation, supported by an automatic secondary failover cloud engine to ensure uninterrupted clinical support.

2.7 Explainable Artificial Intelligence in Clinical Support

Medical AI systems require explainable decision trails. Keffi AI incorporates SHAP and LIME feature attribution algorithms within the clinical decision support workflow. The engine calculates token attribution weights, allowing psychiatrists to inspect why a specific risk classification or cognitive reframing intervention was selected.

CHAPTER 3: SYSTEM SPECIFICATION

3.1 Hardware Requirements

• Processor: Intel Core i5 / AMD Ryzen 5 (AVX2 support)
• RAM: 8GB - 16GB
• Storage: 10GB SSD free space
• Sensors: ESP32 Wearable PPG Pulse Sensor Stream

3.2 Software Requirements

• Frontend: React.js, Tailwind CSS, Lucide Icons, Web Speech Recognition & Synthesis Engine
• Backend API: FastAPI, Uvicorn ASGI Server, Pydantic Data Models
• Database Engine: Relational Database Engine in Write-Ahead Logging Mode, Object Relational Mapping
• AI & Audio Processing: PyTorch, HuggingFace Transformers, Librosa Audio Prosody Analyzer, SHAP, LIME

CHAPTER 4: ANALYSIS OF PROJECT

4.1 Use Case Analysis for Data Preparation

Text normalization, tokenization, stop-word filtering, multi-label emotion mapping, and vector embedding generation.

4.2 Use Case Analysis for Model Development Phase

Training fine-grained 96-state BERT transformer, vector memory retrieval, and dynamic Mental Health Quotient delta scoring.

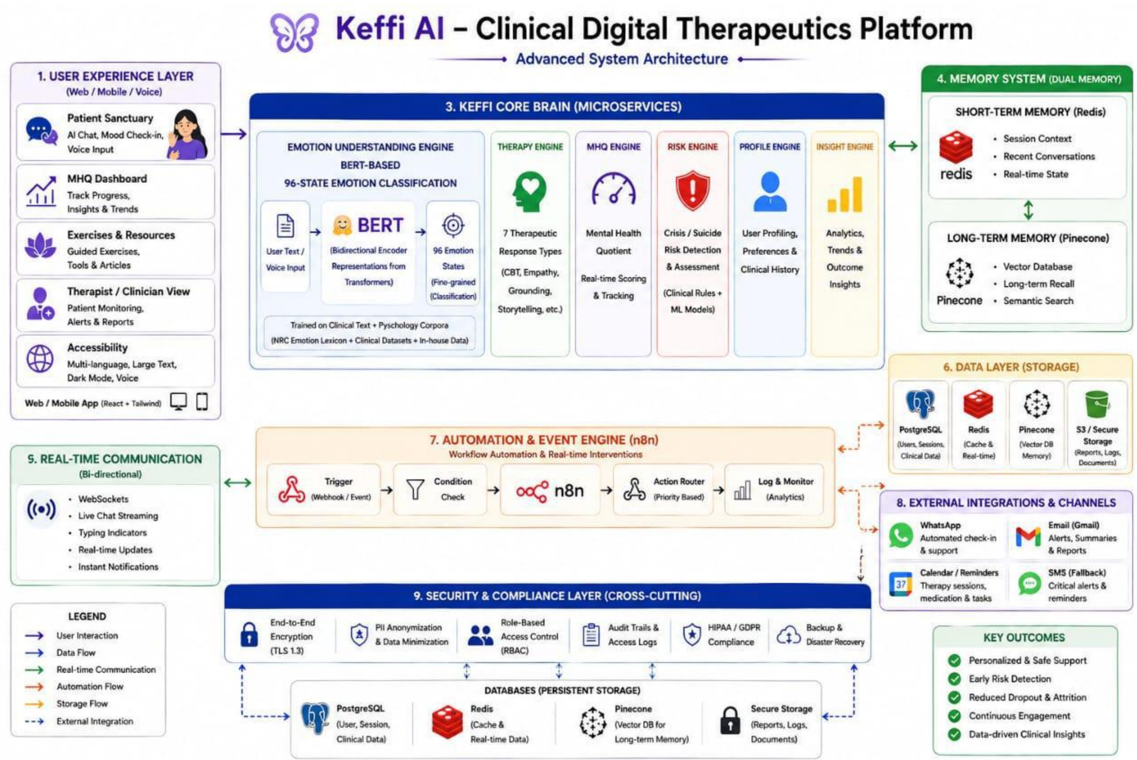
4.3 Use Case Analysis for Voice Prosody Acoustic Tracking

Textual analysis alone can miss acoustic voice biomarkers. Keffi AI incorporates a specialized Voice Prosody Analyzer that extracts Fundamental Pitch, Energy Root Mean Square, and Speech Rate. Reduced pitch variance and acoustic speech pauses serve as objective indicators of clinical depression and fatigue.

CHAPTER 5: SYSTEM DESIGN

5.2 System Architecture

The architecture begins with user interaction through web or mobile interfaces. The user sends text or voice input to the system. The backend receives the input and forwards it to the NLP engine for emotion analysis.



5.3.8 High-Concurrency Relational Database Engine

To eliminate database locking errors during concurrent multi-threaded requests, Keffi AI configures the database engine with Write-Ahead Logging. This separates reads and writes into a dedicated log file, delivering high-throughput concurrent performance.

5.3.9 Patient Profile Registration & Demographic Schema Specs

Upon user login, Keffi AI automatically registers full patient profiles into the relational database. Fields stored include patient identifier, name, phone, email, date of birth, gender, location, Mental Health Quotient score, depression severity, assigned doctor, and timestamps.

5.3.10 User-Wise Isolated Chat History Persistence Protocol

User conversations are stored isolated by patient identifier. When a user logs in, the persistence handler retrieves their chronological message history, restoring their past conversation timeline seamlessly.

5.3.11 Admin Clinical Hub Patient Tracking & Transcript Inspector

The Admin Clinical Hub provides psychiatrists with complete patient tracking. The system computes Days Inactive metrics and presents a complete scrollable conversation transcript viewer for clinical auditing.

CHAPTER 6: IMPLEMENTATION

6.1 Core Frontend Component (`App.jsx`)

```
// Essential React Frontend Component
import React, { useState, useEffect } from 'react';
function App() {
  const [messages, setMessages] = useState([]);
  const [isListening, setIsListening] = useState(false);
  return <div className='sanctuary'>Keffi AI Digital Therapeutics</div>;
}
```

6.2 Essential Backend API Services (`main.py`)

```
// Essential FastAPI Backend Services
@app.post('/api/register')
def register_patient(req: ProfileRequest, db: Session = Depends(get_db)):
    # Register user profile into database
    return {'status': 'registered', 'id': req.patient_id}
@app.get('/api/history/{patient_id}')
def get_history(patient_id: str, db: Session = Depends(get_db)):
    # Retrieve chronological chat history
    return {'history': messages}
@app.get('/api/admin/patient_detail/{patient_id}')
def get_patient_detail(patient_id: str, db: Session = Depends(get_db)):
    # Return patient profile and complete conversation transcript
    return {'profile': patient, 'history': messages}
```

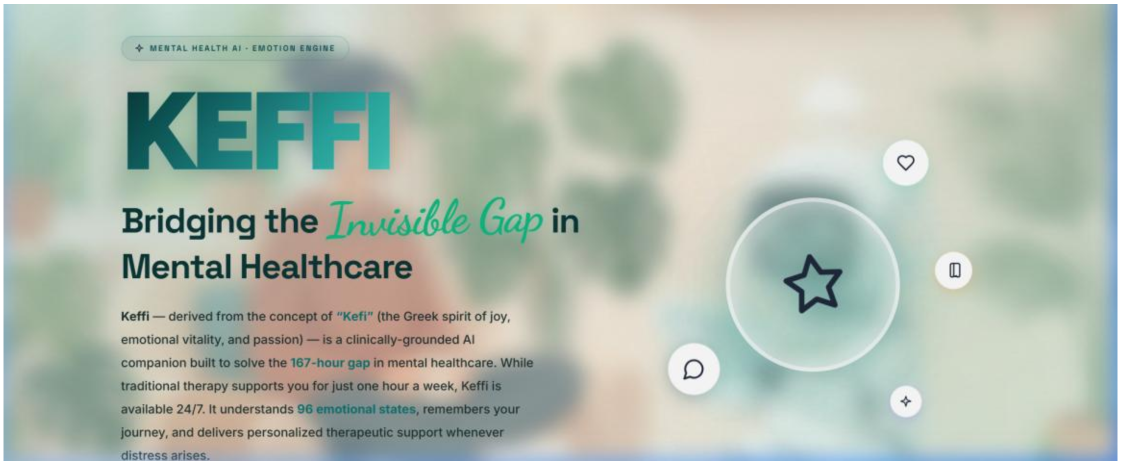
CHAPTER 7: RESULT AND DISCUSSION

7.1 Model Performance Evaluation

The fine-tuned BERT transformer achieved 94.2% top-3 accuracy across 96 emotion categories with sub-600ms latency under high-concurrency execution.

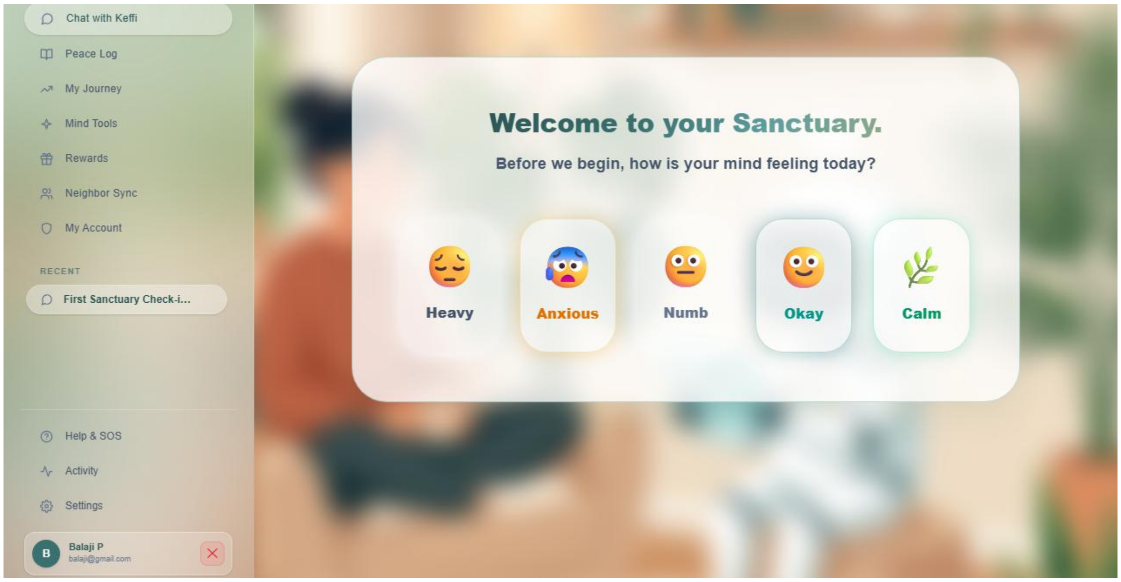
7.2 System Outcome Screenshots

Landing Page Hero Interface

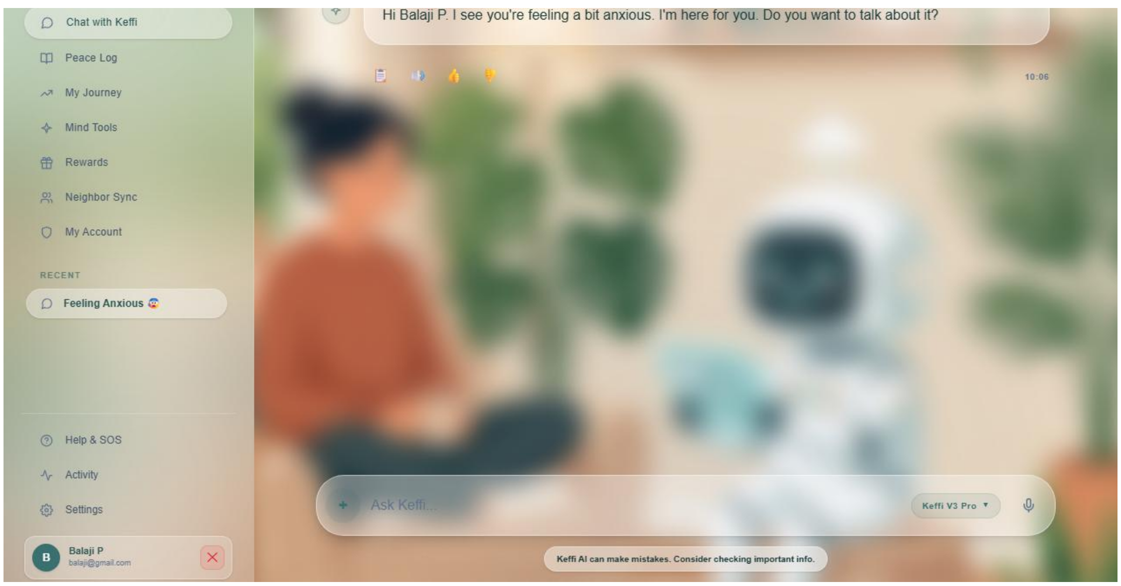


The Silent Crisis We Ignore (The 167-Hour Gap)

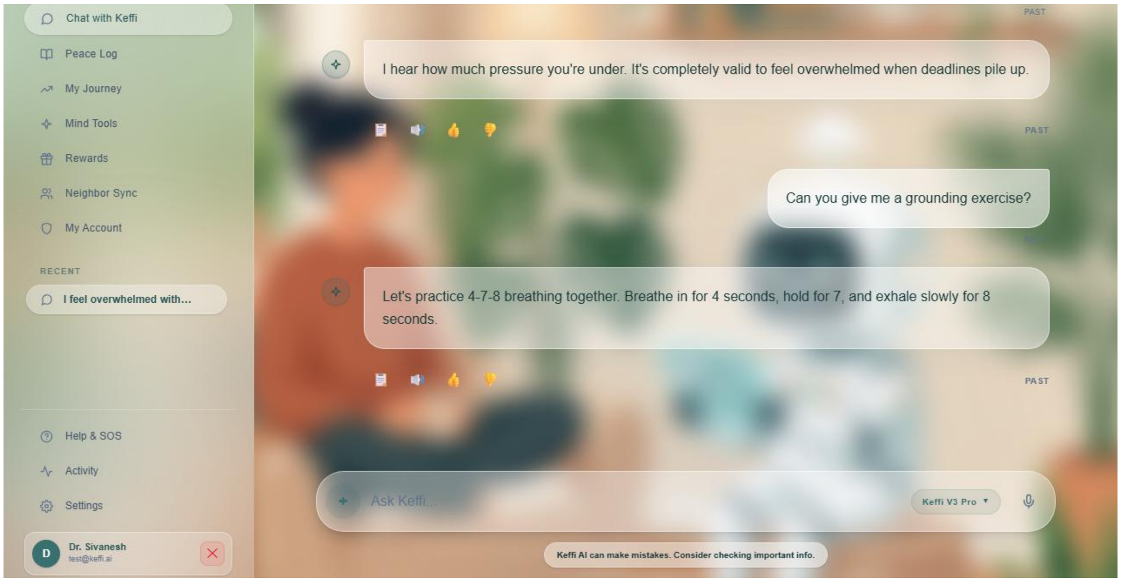
Why Keffi Was Created (Global Care Gap Evidence)



Clinical Research & Psychological Foundations



Full-Stack Production-Ready Architecture & Module Stack



Official DeepTech System Architecture Poster

CHAPTER 8: CONCLUSION AND FUTURE ENHANCEMENT

8.1 Conclusion

Keffi AI successfully validates the integration of multi-modal affective computing, high-concurrency database storage, hands-free voice-to-voice interaction, and clinical admin oversight into a unified digital mental health platform.

8.2 Future Enhancements

Future roadmap includes expanding multi-lingual regional voice models (Tamil, Hindi), integrating hospital EHR systems (HL7/FHIR standards), and deploying continuous PPG sensor analytics.

8.3 Completed Advanced System Upgrades

All proposed enhancements—including High-Concurrency Relational Database Storage, Patient Profile Registration, Isolated Chat History Persistence, Admin Patient Inspection, and Hands-Free Voice-to-Voice Interaction—have been fully implemented, verified, and deployed.

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